

Family Trust Portal - MVP Sprint PRD

Sprint Goal

Build a **functional family portal MVP** that provides secure authentication, basic property management, file uploads, and a foundation for future features. Focus on core functionality that demonstrates value and gets family members using the platform.

Timeline: 4 weeks

Success Criteria: Family members can log in, view properties, upload files, and navigate the basic interface

MVP Features

1. Authentication System ✨

Goal: Secure, user-friendly login for all family members

Magic Link Authentication

- Users enter email address
- Receive secure login link via email
- Click link to authenticate and access dashboard
- Session management with automatic logout

Password Authentication

- Alternative login method for users who prefer passwords
- Secure password requirements and validation
- Password reset functionality via email

NEW: Google OAuth Login

- "Sign in with Google" button on login page
- OAuth integration with Google accounts
- Automatic profile creation from Google account info
- Seamless experience for Gmail users

Role-Based Access

- **Admin:** Full access to all features and settings

- **Member:** Standard family member access
- **Guest:** Limited access (view-only for most content)

Acceptance Criteria:

- ☐ Magic link login works from any device
- ☐ Password login with proper validation
- ☐ Google OAuth integration with profile creation
- ☐ Role-based dashboard views
- ☐ Secure session management
- ☐ Mobile-friendly login experience

2. User Management & Profiles

Goal: Basic user administration and family member profiles

Profile Management

- User profiles with basic information (name, email, role)
- Family relationship designation
- Profile picture upload with automatic optimization
- Contact information and preferences

Admin User Management

- Invite new family members via email
- Assign and modify user roles
- View all family member profiles
- Deactivate accounts if needed

Acceptance Criteria:

- ☐ Profile creation and editing
- ☐ Admin can invite family members
- ☐ Role assignment and modification
- ☐ Profile pictures with image optimization
- ☐ Family relationship tracking

3. Dashboard & Navigation

Goal: Intuitive home base for family portal access

Main Dashboard

- Welcome message with user's name and role
- Quick access cards to main features
- Recent activity feed (basic version)
- Navigation menu for all available features

Responsive Design

- Mobile-first responsive layout
- Touch-friendly navigation on mobile devices
- Consistent experience across desktop, tablet, mobile
- Accessible design with proper contrast and keyboard navigation

Acceptance Criteria:

- ☐ Personalized dashboard based on user role
- ☐ Clear navigation to all features
- ☐ Mobile responsive design
- ☐ Fast loading (< 2 seconds)
- ☐ Accessibility compliance (basic WCAG)

4. Property Management System

Goal: Central repository for family property information

Property Pages

- Create, edit, and view family properties
- Basic property information (name, location, description)
- Property image galleries with optimized loading
- Property status (active, maintenance, unavailable)

Property Details

- Address and location information
- Property guidelines and house rules
- Emergency contact information
- Basic amenities list

Image Management

- Drag-and-drop photo upload for properties
- Automatic image optimization with Sharp
- Multiple image sizes (original, medium, thumbnail)
- Image gallery view with lightbox

Acceptance Criteria:

- ☐ CRUD operations for properties
- ☐ Image upload with Sharp optimization
- ☐ Property detail pages with all information
- ☐ Image galleries with responsive design
- ☐ Admin controls for property management

5. File Upload & Storage

Goal: Secure document and photo storage foundation

File Upload System

- Drag-and-drop file upload interface
- Support for images (JPEG, PNG, WebP) and documents (PDF, DOC)
- File size validation and security checking
- Progress indicators for uploads

Storage Organization

- Files organized by category (property, general, personal)
- Basic file listing and viewing
- Download functionality
- File metadata tracking (uploader, date, size)

Image Processing

- Sharp integration for automatic image optimization
- Generate multiple sizes for responsive display
- WebP conversion for modern browsers
- Thumbnail generation for quick previews

Acceptance Criteria:

- ☐ Drag-and-drop file uploads
- ☐ File type validation and security
- ☐ Image optimization with multiple sizes
- ☐ File organization and metadata
- ☐ Download functionality

6. Basic Admin Panel

Goal: Essential administrative controls for family trustees

User Administration

- View all family members and their roles
- Invite new users with email invitations
- Modify user roles and permissions
- View user activity and login history

Property Administration

- Add, edit, and remove properties
- Manage property images and information
- Set property status and availability
- View property-related activity

File Management

- View all uploaded files across the platform
- File organization and categorization
- Storage usage monitoring
- File access and download logs

Acceptance Criteria:

- ☐ Complete user management interface
- ☐ Property administration tools
- ☐ File management and monitoring
- ☐ Admin-only access controls
- ☐ Activity logging and reporting

Technical Implementation

Tech Stack

- **Frontend:** Next.js 14 with App Router, TypeScript, Tailwind CSS
- **Backend:** Supabase (PostgreSQL, Auth, Storage)
- **UI Components:** shadcn/ui for consistent design
- **Image Processing:** Sharp for optimization
- **Authentication:** Supabase Auth + Google OAuth

Database Schema (MVP)

sql

-- User profiles (extends Supabase auth.users)

```
CREATE TABLE profiles (  
  id UUID REFERENCES auth.users(id) PRIMARY KEY,  
  email TEXT NOT NULL,  
  full_name TEXT,  
  avatar_url TEXT,  
  role TEXT DEFAULT 'member' CHECK (role IN ('admin', 'member', 'guest')),  
  family_relationship TEXT,  
  phone_number TEXT,  
  created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW(),  
  updated_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()  
);
```

-- Properties

```
CREATE TABLE properties (  
  id UUID DEFAULT uuid_generate_v4() PRIMARY KEY,  
  name TEXT NOT NULL,  
  description TEXT,  
  location TEXT,  
  address TEXT,  
  images TEXT[] DEFAULT '{}',  
  amenities TEXT[] DEFAULT '{}',  
  guidelines TEXT,  
  emergency_contacts JSONB DEFAULT '{}',  
  status TEXT DEFAULT 'active' CHECK (status IN ('active', 'maintenance', 'inactive')),  
  created_by UUID REFERENCES auth.users(id),  
  created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW(),  
  updated_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()  
);
```

-- Files and documents

```
CREATE TABLE files (  
  id UUID DEFAULT uuid_generate_v4() PRIMARY KEY,  
  name TEXT NOT NULL,  
  file_path TEXT NOT NULL,  
  file_size BIGINT,  
  content_type TEXT,  
  category TEXT DEFAULT 'general' CHECK (category IN ('property', 'personal', 'general')),  
  property_id UUID REFERENCES properties(id) ON DELETE CASCADE,  
  uploaded_by UUID REFERENCES auth.users(id),  
  created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()  
);
```

-- User invitations

```
CREATE TABLE invitations (  
  id UUID DEFAULT uuid_generate_v4() PRIMARY KEY,  
  email TEXT NOT NULL,  
  role TEXT DEFAULT 'member',  
  invited_by UUID REFERENCES auth.users(id),  
  status TEXT DEFAULT 'pending' CHECK (status IN ('pending', 'accepted', 'expired')),  
  expires_at TIMESTAMP WITH TIME ZONE,  
  created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()  
);
```

Security Requirements

- Row Level Security (RLS) enabled on all tables
- File upload validation and virus scanning
- Role-based access controls throughout
- Secure authentication with proper session management
- Environment variables for all sensitive configuration

Performance Targets

- Page load times < 2 seconds
- Image optimization reduces file sizes by 60-80%
- Mobile performance score > 85 (Lighthouse)
- File uploads support up to 50MB
- Concurrent users: 20+ family members

MVP Success Metrics

Technical Metrics

- ☐ All authentication methods working (magic link, password, Google)
- ☐ Property pages loading in < 2 seconds
- ☐ File uploads completing successfully 95%+ of time
- ☐ Mobile responsiveness on all major devices
- ☐ Zero critical security vulnerabilities

User Adoption Metrics

- ☐ 80%+ of invited family members complete registration
- ☐ Each family member logs in at least once within first week

- ☐ At least 50% of family members upload a photo or document
- ☐ 100% of family properties have basic information added
- ☐ Admin successfully manages users and properties

Quality Metrics

- ☐ User feedback score > 4/5 for ease of use
- ☐ Zero data loss incidents
- ☐ Login success rate > 98%
- ☐ Mobile usability rated as "good" by family members
- ☐ No major bugs reported in core functionality

MVP Limitations (Intentional)

- **No Calendar/Booking:** Property scheduling deferred to future sprint
- **No AI Chat:** Document AI features planned for later
- **No Google Drive Integration:** File storage via platform only for now
- **Basic Admin Panel:** Advanced analytics and reporting later
- **Limited Social Features:** Focus on core functionality first

Definition of MVP Done

The MVP is complete when:

- ☐ All three authentication methods work reliably
- ☐ Family members can view and manage properties
- ☐ File upload and image optimization working
- ☐ Admin can manage users and content
- ☐ Mobile responsive design implemented
- ☐ Basic security measures in place
- ☐ Deployed and accessible to family members
- ☐ Family feedback collected for next sprint planning

Next Sprint Planning

After MVP completion:

1. **Collect Family Feedback:** Survey on most wanted features
2. **Prioritize Enhancements:** Based on family input and usage data
3. **Plan Sprint 2:** Focus on highest-value additions (likely calendar/booking)

4. **Technical Debt:** Address any MVP shortcuts or performance issues

Risk Mitigation

Technical Risks

- **Google OAuth Complexity:** Start with simple implementation, enhance later
- **Image Processing Performance:** Monitor Sharp processing times, optimize if needed
- **Mobile Performance:** Test on family members' actual devices early

User Adoption Risks

- **Learning Curve:** Provide simple onboarding flow and help documentation
- **Login Confusion:** Clear instructions for all three authentication methods
- **Mobile Usability:** Prioritize mobile experience since family uses phones primarily

Security Risks

- **File Upload Security:** Implement strict validation and malware scanning
- **Role Permissions:** Thoroughly test admin vs member access controls
- **Session Security:** Proper logout and session expiration handling

This MVP provides a solid foundation that demonstrates value while setting up the architecture for future enhancements based on family feedback and usage patterns.